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ME 40: Thermodynamics

Fall 2026
Format: In person
3 Units

Course description: This course introduces the scientific principles that deal with energy conversion among different forms, such as heat, work, internal energy, electrical energy, and chemical energy. The physical science of heat and temperature, and their relationship to energy and work, are analyzed on the basis of the fundamental thermodynamic laws. These principles are applied to various practical systems, including heat engines, refrigeration cycles, and air conditioning systems.

Learning objectives:

I. *Knowledge:* Write the first and second laws of thermodynamics.

II. *Comprehension*: Describe the first and second laws in the student's own language.

III. *Application*: Solve simple single-answer problems using the first law.

IV. *Analysis*: Solve problems requiring both the first and second laws.

V. *Synthesis*: Develop schemes to improve the efficiency or performance of thermodynamic systems, e.g., compressors, turbines, heat pumps, etc.

VI. *Evaluation*: Determine and describe second law fallacies in proposed power cycles.

VII. *Evaluation*: Judge when classical thermodynamics is not the appropriate analysis tool.

VIII. *Evaluation*: Find and correct errors in the student's own solutions and in those of others.

IX. *Analysis and Evaluation*: Search appropriate databases and the literature to find required thermodynamic data, and if the data are not available the student can select appropriate procedures and predict the values of the data.

X. *Knowledge*: Briefly outline the history of the field of thermodynamics.

Instructor: Prof. Claudio Hail; email: cuhail@berkeley.edu

Office Hours and Location: Monday 3:00 pm -4:00 pm, 6173 Etcheverry Hall.

If there is demand, additional office hours will be added. Please contact the instructor.

Teaching Support Staff

Head GSI	GSI
Cassidy McCormick,	Jessica Wu,
cjmccormick@berkeley.edu	jywu@berkeley.edu
Office Hours: Thur, 2:30 pm – 3:30 pm,	Office Hours: Mo,10:00 am – 11:00 am,
Hesse Hall, “fishbowl”.	Hesse Hall, “fishbowl”.

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Reader	Reader
Ninad Atale,	Takenori Tsuchida,
ninad.atale@berkeley.edu	takenori_tsuchida@berkeley.edu

Emails to course staff: Begin subject line with “ME40: …”

Messages on Canvas are not regularly monitored. Email is best for communication.

Required text: Çengel and Boles, *Thermodynamics: An Engineering Approach*, **10th Edition** Other recent editions may be acceptable (e.g., 7th, 8th, or 9th), but it is your responsibility to determine the correct homework problems and readings if they do not match up.

Lecture: Monday, Wednesday, Friday at 2:00 pm-3:00 pm in Hearst Mining 390.

There may also be a video playlist for each lecture. Please watch the video playlist before attending live lecture or section.

Discussions: 101: Tuesday, 5:00 pm – 6:00 pm in Stanley 106, Jessica Wu

102: Thursday, 1:00 pm – 2:00 pm in Etcheverry 3111, Cassidy McCormick

103: Friday, 12:00 pm – 1:00 pm in Etcheverry 3107, Cassidy McCormick

Website: All teaching material is available for download from this link:

https://bcourses.berkeley.edu/courses/1557315

Grading:

Midterm 1	Worth 40 points
Midterm 2	Worth 40 points
Final Exam	Worth 70 points
One Project	Worth 20 points
Homework	Worth 30 points (there are 12 homework assignments)
Total of 200 points	In a typical semester, >70% of students taking the course for a letter grade will receive a B- or higher.

Attendance: Attendance at lectures and discussions is expected.

Homework: We will use PrairieLearn for homework submission and grading. For regular homework questions retakes are allowed with no penalty for number of attempts. For high stakes questions no retakes are allowed. For full credit a document that shows your work must be submitted. Depending on the content, we may also ask for handwritten homework to be submitted, and in this case, we may only grade a subset of the problems and retakes are not allowed. Late submission of the homework will reduce credit as shown in Tab. 1. Drop policy: The lowest scoring homework will be dropped from the final grade calculation.

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Midterm 1 & 2

Duration: 50 minutes. In-person, closed book and notes. Programmable calculator (non-transmittable) and 3 letter-sized sheets of course-provided summary (8.5” x 11”, double-sided) for the exam is permitted. There is strictly no writing, highlighting or markings allowed on the sheets.

Midterm 1 will take place on 10/5/26 during class hours.

Midterm 2 will take place on 11/9/26 during class hours.

Final:

Duration: 3 hours. Closed book and notes. Programmable calculator (non-transmittable) and3 letter-sized sheets of course-provided summary (8.5” x 11”, double-sided) for the exam is permitted. There is strictly no writing, highlighting or markings allowed on the sheets.

The final exam will take place on 12/17/26 at 3–6 pm.

Project:

The project is a larger assignment intended to combine ideas from the course in interesting ways. You are required to work in groups of three. Projects are graded on correctness, quality of the presentation and understanding. Creativity is encouraged. See Lecture 1a slides for posting and due dates.

Exam recovery:

It is possible to recover points lost in the midterm through improved performance on the final exam. This is consistent with the principle of testing for proficiency. Your midterm percentages will be calculated as:

$$M_{1,final} = \max(M_{1,original} \cdot 0.6 + Final \cdot 0.4, M_{1,original})$$

$$M_{2,final} = \max\left(M_{2,original} \cdot 0.6 + Final \cdot 0.4, M_{2,original}\right)$$

For example, if you received 80% on midterm 1, 50% on midterm 2, and 75% on the final, your midterm scores would be:

$$M_{1,final} = \max(0.8 \cdot 0.6 + 0.75 \cdot 0.4, 0.8) = 0.8,$$

$$M_{2,final} = \max(0.5 \cdot 0.6 + 0.75 \cdot 0.4, 0.5) = 0.6.$$

Regrades:

Any serious concerns about grading should be addressed to the instructor, not the GSIs or Reader, within seven days of receiving the graded homework, quiz, or exam. Please include a brief written explanation of your concern. Re-graded scores may increase, decrease, or remain the same. The instructor reserves the right to regrade the other problems on the homework, quiz, or exam too.

Absences, and makeups:

Lectures: Ask a classmate for the notes.

Exams: Missing an exam will result in a zero grade for that exam. Exceptions are made for severe medical (doctor’s note required) or family emergencies. When granted, makeup exams may be oral or written.

Other expectations:

Questions are encouraged.

Silence your cellphones.

Treat your colleagues, Reader, GSIs, and Instructor with respect.

No food or drinks, except for water.

Cheating:	Please review the Berkeley Campus Code of Student Conduct.
Use of AI:	The use of generative AI tools is discouraged in this course. Assignments are designed to foster your critical thinking, problem solving and subject matter understanding. Homework submissions may be checked for AI-use. Submitting AI-generated material as your original work violates the course academic integrity policy and may result in disciplinary action.
Student with disabilities:	If you require course accommodations due to a physical, emotional, or learning disability, contact UC Berkeley's Disabled Students' Program (DSP). Notify the instructor and GSI through course email of the accommodations you would like to use. You must have a Letter of Accommodation on file with UC Berkeley to have accommodations made in the course. DSP accommodations for exams: If more than “Reduced Distraction” and “Extra time” are required, please apply with the DSP Proctoring Office to take the exam at their facility. Do this as early as possible. We have an Alternative Testing Agreement in place. DSP accommodations for homework: If “Extensions to Assignment Deadlines” is required, please email the instructor to inquire about the assignment deadline.
Other:	Due to travel, several lectures may be pre-recorded and uploaded or someone else may deliver the lecture. You will be notified well in advance of those dates.
Parting thought:	Grades and penalties are not the purpose of this course. We genuinely just want you to learn. All of us are very excited to be teaching ME40 this semester and look forward to meeting such a large and enthusiastic group of students!

Day	**Credit**
Due date / Sunday	100%
1 day late / Monday	95%
2 days late / Tuesday	90%
3 days late / Wednesday	85%
4 days late / Thursday	80%
5 days late / Friday	70%
6 days late / Saturday	60%
7 or more days late until the end of module	50%

***Table 1**. Homework credit for late submission*